

Questions with Answer Keys

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Q1: 24 Feb (Shift 1) - Single Correct

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = 2x - 1$ and $g: \mathbb{R} - \{1\} \rightarrow \mathbb{R}$ be defined as $g(x) = \frac{x - \frac{1}{2}}{x - 1}$. Then the composition function $f(g(x))$ is :

- (1) both one-one and onto
- (2) onto but not one-one
- (3) neither one-one nor onto
- (4) one-one but not onto

Q2: 24 Feb (Shift 2) - Numerical

If $a + \alpha = 1, b + \beta = 2$ and $af(x) + \alpha f\left(\frac{1}{x}\right) = bx + \frac{\beta}{x}, x \neq 0$, then the value of the expression $\frac{f(x) + f\left(\frac{1}{x}\right)}{x + \frac{1}{x}}$ is

Q3: 25 Feb (Shift 1) - Single Correct

Let $f, g: \mathbb{N} \rightarrow \mathbb{N}$ such that $f(n+1) = f(n) + f(a) \quad \forall n \in \mathbb{N}$ and g be any arbitrary function. Which of the following statements is NOT true?

- (1) f is one-one
- (2) If $f \circ g$ is one-one, then g is one-one
- (3) If g is onto, then $f \circ g$ is one-one
- (4) If f is onto, then $f(n) = n \forall n \in \mathbb{N}$

Q4: 25 Feb (Shift 2) - Single Correct

Let x denote the total number of one-one functions from a set A with 3 elements to a set B with 5 elements and y denote the total number of one-one functions from the set A to the set $A \times B$. Then:

- (1) $y = 273x$
- (2) $2y = 91x$

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(3) $y = 91x$

(4) $2y = 273x$

Q5: 25 Feb (Shift 2) - Single Correct

A function $f(x)$ is given by $f(x) = \frac{5^x}{5^x + 5}$, then the sum of the series

$f\left(\frac{1}{20}\right) + f\left(\frac{2}{20}\right) + f\left(\frac{3}{20}\right) + \dots + f\left(\frac{39}{20}\right)$ is equal to:

(1) $\frac{19}{2}$

(2) $\frac{49}{2}$

(3) $\frac{39}{2}$

(4) $\frac{29}{2}$

Q6: 26 Feb (Shift 1) - Single Correct

Let $R = \{P, Q\} \mid P \text{ and } Q \text{ are at the same distance from the origin}\}$ be a relation, then the equivalence class of $(1, -1)$ is the set:

(1) $S = \{(x, y) \mid x^2 + y^2 = 1\}$

(2) $S = \{(x, y) \mid x^2 + y^2 = 4\}$

(3) $S = \{(x, y) \mid x^2 + y^2 = \sqrt{2}\}$

(4) $S = \{(x, y) \mid x^2 + y^2 = 2\}$

Q7: 26 Feb (Shift 2) - Single Correct

Let $A = \{1, 2, 3, \dots, 10\}$ and $f: A \rightarrow A$ be defined as

$$f(k) = \begin{cases} k+1 & \text{if } k \text{ is odd} \\ k & \text{if } k \text{ is even} \end{cases}$$

Then the number of possible functions $g: A \rightarrow A$ such that $\text{gof} = f$ is:

(1) 10^5

(2) ${}^{10}C_5$

(3) 5^5

(4) $5!$

Q8: 26 Feb (Shift 2) - Single Correct

Let $f(x) = \sin^{-1} x$ and $g(x) = \frac{x^2 - x - 2}{2x^2 - x - 6}$. If $g(2) = \lim_{x \rightarrow 2} g(x)$, then the

domain of the function fog is :

(1) $(-\infty, -2] \cup \left[-\frac{4}{3}, \infty\right)$

(2) $(-\infty, -1] \cup [2, \infty)$

(3) $(-\infty, -2] \cup [-1, \infty)$

(4) $(-\infty, -2] \cup \left[-\frac{3}{2}, \infty\right)$

Answer Key**Q1 (4)****Q2 (2)****Q3 (3)****Q4 (2)****Q5 (3)****Q6 (4)****Q7 (1)****Q8 (1)**

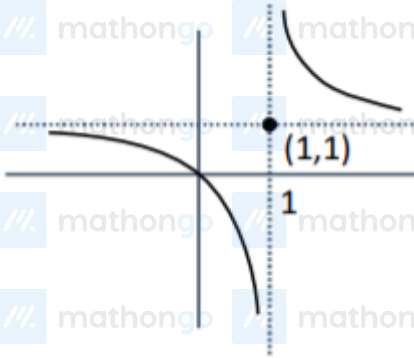
Q1 (4)

$$f(g(x)) = 2g(x) - 1$$

$$= 2 \frac{\left(x - \frac{1}{2}\right)}{x-1} = \frac{x}{x-1}$$

$$f(g(x)) = 1 + \frac{1}{x-1}$$

one-one, into



Q2 (2)

$$af(x) + \alpha f\left(\frac{1}{x}\right) = bx + \frac{\beta}{x} \dots\dots\dots$$

$$x \rightarrow \frac{1}{x}$$

$$af\left(\frac{1}{x}\right) + af(x) = \frac{b}{x} + \beta x$$

.....

(i) + (ii)

$$(a + \alpha) \left[f(x) + f\left(\frac{1}{x}\right) \right] = \left(x + \frac{1}{x} \right) (b + \beta)$$

$$\frac{f(x) + f\left(\frac{1}{x}\right)}{x + \frac{1}{x}} = \frac{2}{1} = 2$$

Q3 (3)

$$f(n+1) = f(n) + 1$$

$$f(2) = 2f(1)$$

$$f(3) = 3f(1)$$

$$f(4) = 4f(1)$$

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Hints and Solutions

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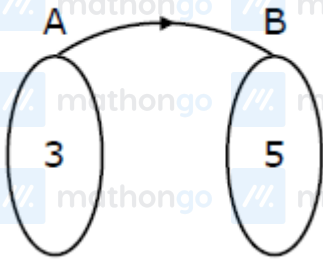
$$f(n) = nf(1)$$

$f(x)$ is one-one

Q4 (2)

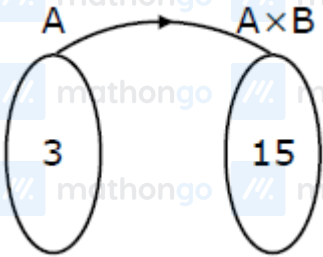
Number of elements in $A = 3$

Number of elements in $B = 5$ Number of elements in $A \times B = 15$



Number of one-one function $x = 5 \times 4 \times 3$

$$x = 60$$



Number of one-one function $y = 15 \times 14 \times 13$

$$y = 15 \times 4 \times \frac{14}{4} \times 13$$

$$y = 60 \times \frac{7}{2} \times 13$$

$$2y = (13)(7x)$$

$$2y = 91x$$

Q5 (3)

$$f(x) = \frac{5^x}{5^x + 5} \dots (i)$$

$$f(2-x) = \frac{5^{2-x}}{5^{2-x} + 5}$$

Hints and Solutions

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$$f(2-x) = \frac{5}{5^x+5} \dots\dots$$

Adding equation (i) and(ii)

$$f(x) + f(2-x) = 1$$

$$f\left(\frac{1}{20}\right) + f\left(\frac{39}{20}\right) = 1$$

$$f\left(\frac{2}{20}\right) + f\left(\frac{38}{20}\right) = 1$$

$$f\left(\frac{19}{20}\right) + f\left(\frac{21}{20}\right) = 1$$

$$\text{and } f\left(\frac{20}{20}\right) = f(1) = \frac{1}{2}$$

$$\Rightarrow 19 + \frac{1}{2} \Rightarrow \frac{39}{2}$$

Q6 (4)

$$P(a, b), Q(c, d), PO = QO$$

$$\Rightarrow a^2 + b^2 = c^2 + d^2$$

$$R(x, y) \quad s = (1, -1) \Rightarrow RO = SO$$

(\$\therefore\$ equivalence class)

$$x^2 + y^2 = 2$$

Q7 (1)

$$g(f(x)) = f(x)$$

$$\Rightarrow g(x) = x, \text{ when } x \text{ is even}$$

5 elements in A can be mapped to any 10

$$\text{So, } 10^5 \times 1 = 10^5$$

Q8 (1)

$$g(2) = \lim_{x \rightarrow 2} \frac{(x-2)(x+1)}{(2x+3)(x-2)} = \frac{3}{7}$$

$$\text{For domain of fog (x) } \left| \frac{x^2-x-2}{2x^2-x-6} \right| \leq 1 \Rightarrow (x+1)^2 \leq (2x+3)^2 \Rightarrow 3x^2 + 10x + 8 \geq 0$$

$$\Rightarrow (3x+4)(x+2) \geq 0$$

$$x \in (-\infty, -2] \cup \left(-\frac{4}{3}, \infty\right]$$